Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: GJC Dairy #1

Physical address of dairy:

Number and Street	City	County	Zip Code
2401 Newman RD	Turlock	Stanislaus	95380

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 11/01/1985

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X058-X014-X018-XXXX

B. OPERATORS

Vieira, Jim

Operator name: Vieira, Jim

Telephone no.: (209) 668-2244 Landline, (209) 652-6980 Cellular

Mailing Address Number and Street	City	State	Zip Code
P.O. Box 2397	Turlock	CA	95381

This operator is responsible for paying permit fees.

C. OWNERS

Vieira, Jim

Legal owner name: Vieira, Jim

Telephone no.: (209) 668-2244 Landline, (209) 652-6980 Cellular

Mailing Address Number and Street	City	State	Zip Code
P.O. Box 2397	Turlock	CA	95381

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	50	250	0	0
Number under roof	600	96	0	0	150	0
Maximum number	600	96	50	250	150	0
Average number	600	96	50	250	150	0
Avg live weight (lbs)	1,400	1,450	900	600		

Predominant milk cow breed: Holstein

Average milk production: 73 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 20,317.32 *tons per reporting period*

Total nitrogen from manure: 257,645.09 *lbs per reporting period*

After ammonia losses (30% loss applied): 180,351.56 *lbs per reporting period*

Total phosphorus from manure: 42,247.23 *lbs per reporting period*

Total potassium from manure: 116,088.14 *lbs per reporting period*

Total salt from manure: 304,585.20 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 4,066,000 *gallons*

Total nitrogen generated: 42,691.00 *lbs*

Total phosphorus generated: 13,188.12 *lbs*

Total potassium generated: 53,245.94 *lbs*

Total salt generated: 219,640.40 *lbs*

	4,066,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	4,066,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
TID Ceres Main	Surface water
TID Lateral 4	Surface water
TID Lateral 5	Surface water
TID Lateral 6	Ground water
TID Turlock Main	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
04/14/2023	Solid commercial fertilizer *Potash*	20.34 *ton*	Dry-weight	0.1	0.000000	0.000000	60.000000	0.000000
06/13/2023	Solid commercial fertilizer *UN32*	2.70 *ton*	Dry-weight	0.1	32.000000	0.000000	0.000000	0.000000
08/09/2023	Solid commercial fertilizer *NH3*	24.64 *ton*	Dry-weight	0.1	82.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	42,095.46	0.00	24,383.59	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	42,095.46	0.00	24,383.59	0.00

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Dad's	48	48	1	none	X022-X044-X003-XXXX
Hirdes East	90	90	1	process wastewater	X058-X019-X002-XXXX X058-X019-X003-XXXX
Hirdes West	188	188	2	manure	X058-X019-X007-XXXX X058-X020-X001-XXXX X058-X020-X002-XXXX X058-X020-X005-XXXX
Home	8	8	2	process wastewater	X058-X014-X014-XXXX
Jakes	8	8	2	process wastewater	X058-X014-X013-XXXX
Lander - Duane Almeida	16	16	1	manure	X045-X100-X022-XXXX
Scale	8	8	2	process wastewater	X058-X014-X018-XXXX
Tony's	37	37	2	manure	X057-X012-X018-XXXX
Triangle	21	21	2	process wastewater	X058-X014-X018-XXXX X058-X014-X019-XXXX
Verduga	16	16	1	manure	X024-X045-X022-XXXX
Washington - Sandoval	18	18	1	none	X044-X013-X015-XXXX
Totals for areas that were used for application	392	392	15		
Totals for areas that were not used for application	66	66	2		
Land application area totals	458	458	17		

B. CROPS AND HARVESTS

Dad's

Field name: Dad's

Dad's

01/28/2021: Almond, in shell

Crop: Almond, in shell Acres planted: 48 Plant date: 01/28/2021

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
12/31/2023	0.10 *ton*	As-is		0.1	0.00	0.00	0.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	1.00	1.00	1.00	1.00	1.00
Total actual harvest content	0.00	0.00	0.00	0.00	0.00

Hirdes East

Field name: Hirdes East

10/20/2020: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 90 Plant date: 10/20/2020

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/15/2023	724.50 *ton*	As-is		10.2	28,100.00	1,810.00	17,500.00		11.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	7.50	450.00	40.50	315.00	0.00
Total actual harvest content	8.05	452.41	29.14	281.75	1,691.56

Hirdes West

Field name: Hirdes West

Hirdes West

11/22/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 188 Plant date: 11/22/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/25/2023	4,248.41 *ton*	As-is		70.0	3,200.00	800.00	5,000.00		7.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	22.60	144.63	36.16	225.98	949.11

06/12/2023: Corn, silage

Crop: Corn, silage Acres planted: 188 Plant date: 06/12/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/11/2023	5,417.97 *ton*	As-is		68.0	3,900.00	600.00	3,300.00		5.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	28.82	224.79	34.58	190.21	940.65

Home

Field name: Home

10/27/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 8 Plant date: 10/27/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/17/2023	98.21 *ton*	As-is		70.0	6,200.00	1,100.00	10,300.00		8.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	12.28	152.23	27.01	252.89	648.19

Home

06/16/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 8 Plant date: 06/16/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/09/2023	148.30 *ton*	As-is		55.2	11,600.00	1,700.00	22,000.00		16.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	154.00	23.80	168.00	0.00
Total actual harvest content	18.54	430.07	63.03	815.65	2,657.54

Jakes

Field name: Jakes

10/24/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 8 Plant date: 10/24/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/17/2023	84.00 *ton*	As-is		70.0	9,100.00	1,300.00	14,000.00		14.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	10.50	191.10	27.30	294.00	882.00

06/12/2023: Corn, silage

Crop: Corn, silage Acres planted: 8 Plant date: 06/12/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/11/2023	169.00 *ton*	As-is		57.1	6,900.00	1,300.00	9,300.00		20.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	21.13	291.53	54.93	392.93	3,770.05

Lander - Duane Almeida

Field name: Lander - Duane Almeida

10/22/2020: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 16 Plant date: 10/22/2020

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/15/2023	108.75 *ton*	As-is		9.5	26,200.00	1,650.00	16,500.00		15.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	7.50	450.00	40.50	315.00	0.00
Total actual harvest content	6.80	356.16	22.43	224.30	1,906.86

Scale

Field name: Scale

10/27/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 8 Plant date: 10/27/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/17/2023	148.00 *ton*	As-is		70.0	6,200.00	1,100.00	10,300.00		8.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	18.50	229.40	40.70	381.10	976.80

06/16/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 8 Plant date: 06/16/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/09/2023	131.96 *ton*	As-is		55.2	11,600.00	1,700.00	22,000.00		16.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	154.00	23.80	168.00	0.00
Total actual harvest content	16.50	382.68	56.08	725.78	2,364.72

Tony's

Field name: Tony's

11/14/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 37 Plant date: 11/14/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	842.91 *ton*	As-is		70.0	4,200.00	800.00	5,600.00		7.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	22.78	191.36	36.45	255.15	1,052.50

06/13/2023: Corn, silage

Crop: Corn, silage Acres planted: 37 Plant date: 06/13/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/12/2023	1,049.27 *ton*	As-is		72.3	4,000.00	800.00	3,600.00		5.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	28.36	226.87	45.37	204.18	879.80

Triangle

Field name: Triangle

11/14/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 18 Plant date: 11/14/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	340.00 *ton*	As-is		70.0	4,100.00	800.00	5,900.00		7.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	18.89	154.89	30.22	222.89	861.33

Triangle

06/16/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 18 Plant date: 06/16/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/09/2023	244.00 *ton*	As-is		59.9	9,600.00	1,400.00	17,900.00		14.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	154.00	23.80	168.00	0.00
Total actual harvest content	13.56	260.27	37.96	485.29	1,608.99

Verduga

Field name: Verduga

10/22/2020: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 16 Plant date: 10/22/2020

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/15/2023	118.25 *ton*	As-is		9.5	26,200.00	1,650.00	16,500.00		15.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	7.50	450.00	40.50	315.00	0.00
Total actual harvest content	7.39	387.27	24.39	243.89	2,073.44

Washington - Sandoval

Field name: Washington - Sandoval

Washington - Sandoval

01/28/2016: Almond, in shell

Crop: Almond, in shell Acres planted: 18 Plant date: 01/28/2016

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/30/2023	74.00 *ton*	As-is		7.8	24,200.00	1,700.00	15,900.00		11.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	5.00	215.00	36.50	232.50	0.00
Total actual harvest content	4.11	198.98	13.98	130.73	886.96

NUTRIENT BUDGET

A. LAND APPLICATIONS

Dad's - 01/28/2021: Almond, in shell

Field name: Dad's

Crop: Almond, in shell

Plant date: 01/28/2021

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	11.72	2,592,000.00 *gal*
Application event totals		0.00	0.00	0.00	11.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	11.72	2,592,000.00 *gal*
Application event totals		0.00	0.00	0.00	11.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	11.72	2,592,000.00 *gal*
Application event totals		0.00	0.00	0.00	11.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	11.72	2,592,000.00 *gal*
Application event totals		0.00	0.00	0.00	11.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	11.72	2,592,000.00 *gal*
Application event totals		0.00	0.00	0.00	11.72	

Dad's - 01/28/2021: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	11.72	2,592,000.00 *gal*
Application event totals		0.00	0.00	0.00	11.72	

06/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	11.72	2,592,000.00 *gal*
Application event totals		0.00	0.00	0.00	11.72	

06/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	11.72	2,592,000.00 *gal*
Application event totals		0.00	0.00	0.00	11.72	

07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	11.72	2,592,000.00 *gal*
Application event totals		0.00	0.00	0.00	11.72	

07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	11.72	2,592,000.00 *gal*
Application event totals		0.00	0.00	0.00	11.72	

07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	11.72	2,592,000.00 *gal*
Application event totals		0.00	0.00	0.00	11.72	

Dad's - 01/28/2021: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	11.72	2,592,000.00 *gal*
Application event totals		0.00	0.00	0.00	11.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	11.72	2,592,000.00 *gal*
Application event totals		0.00	0.00	0.00	11.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	11.72	2,592,000.00 *gal*
Application event totals		0.00	0.00	0.00	11.72	

Hirdes East - 10/20/2020: Alfalfa, hay

Field name: Hirdes East

Crop: Alfalfa, hay

Plant date: 10/20/2020

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/17/2023	Towed tank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater sludge	199.71	54.51	175.89	723.72	988,000.00 *gal*
Application event totals		199.71	54.51	175.89	723.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/11/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Potash	Solid commercial fertilizer	0.00	0.00	200.10	0.00	
Application event totals		0.00	0.00	200.10	0.00	

Hirdes East - 10/20/2020: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/12/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Potash	Solid commercial fertilizer	0.00	0.00	200.10	0.00	
Application event totals		0.00	0.00	200.10	0.00	

04/13/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Potash	Solid commercial fertilizer	0.00	0.00	200.10	0.00	
Application event totals		0.00	0.00	200.10	0.00	

04/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	41.01	17,010,000.00 *gal*
Application event totals		0.00	0.00	0.00	41.01	

05/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	35.15	14,580,000.00 *gal*
Application event totals		0.00	0.00	0.00	35.15	

06/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	35.15	14,580,000.00 *gal*
Application event totals		0.00	0.00	0.00	35.15	

07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	33.20	13,770,000.00 *gal*
Application event totals		0.00	0.00	0.00	33.20	

Hirdes East - 10/20/2020: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	31.24	12,960,000.00 *gal*
Application event totals		0.00	0.00	0.00	31.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	35.15	14,580,000.00 *gal*
Application event totals		0.00	0.00	0.00	35.15	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	35.15	14,580,000.00 *gal*
Application event totals		0.00	0.00	0.00	35.15	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	34.17	14,175,000.00 *gal*
Application event totals		0.00	0.00	0.00	34.17	

Hirdes West - 11/22/2022: Oats, silage-soft dough

Field name: Hirdes West

Crop: Oats, silage-soft dough

Plant date: 11/22/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/14/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator Solids	Separator solids	200.23	45.13	109.97	1,156.86	2,500.00 *ton*
Application event totals		200.23	45.13	109.97	1,156.86	

Hirdes West - 11/22/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/31/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	0.00	27,540,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	32.72	28,350,000.00 *gal*
Application event totals		0.00	0.00	0.00	32.72	

Hirdes West - 06/12/2023: Corn, silage

Field name: Hirdes West

Crop: Corn, silage

Plant date: 06/12/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/27/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator Solids	Separator solids	123.87	30.65	104.48	0.00	2,170.00 *ton*
Application event totals		123.87	30.65	104.48	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	33.65	29,160,000.00 *gal*
Application event totals		0.00	0.00	0.00	33.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
NH3	Solid commercial fertilizer	59.86	0.00	0.00	0.00	
Application event totals		59.86	0.00	0.00	0.00	

Hirdes West - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	32.41	28,080,000.00 *gal*
Application event totals		0.00	0.00	0.00	32.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
NH3	Solid commercial fertilizer	59.86	0.00	0.00	0.00	
Application event totals		59.86	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	31.16	27,000,000.00 *gal*
Application event totals		0.00	0.00	0.00	31.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/06/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
NH3	Solid commercial fertilizer	59.86	0.00	0.00	0.00	
Application event totals		59.86	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	29.91	25,920,000.00 *gal*
Application event totals		0.00	0.00	0.00	29.91	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	28.67	24,840,000.00 *gal*
Application event totals		0.00	0.00	0.00	28.67	

Hirdes West - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	27.42	23,760,000.00 *gal*
Application event totals		0.00	0.00	0.00	27.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	28.67	24,840,000.00 *gal*
Application event totals		0.00	0.00	0.00	28.67	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	26.80	23,220,000.00 *gal*
Application event totals		0.00	0.00	0.00	26.80	

Home - 10/27/2022: Oats, silage-soft dough

Field name: Home

Crop: Oats, silage-soft dough

Plant date: 10/27/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/06/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	101.50	33.91	114.91	461.90	108,000.00 *gal*
TID Ceres Main	Surface water	1.39	0.00	0.00	0.00	810,000.00 *gal*
Application event totals		102.89	33.91	114.91	461.90	

Home - 10/27/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	110.52	37.06	179.13	743.54	108,000.00 *gal*
TID Ceres Main	Surface water	0.00	0.00	0.00	8.45	675,000.00 *gal*
Application event totals		110.52	37.06	179.13	751.99	

Home - 06/16/2023: Sudangrass, silage

Field name: Home

Crop: Sudangrass, silage

Plant date: 06/16/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Ceres Main	Surface water	0.00	0.00	0.00	13.52	1,080,000.00 *gal*
Application event totals		0.00	0.00	0.00	13.52	

07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	128.94	43.24	208.98	867.46	126,000.00 *gal*
TID Ceres Main	Surface water	0.00	0.00	0.00	11.83	945,000.00 *gal*
Application event totals		128.94	43.24	208.98	879.29	

07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Ceres Main	Surface water	0.00	0.00	0.00	11.83	945,000.00 *gal*
Application event totals		0.00	0.00	0.00	11.83	

Home - 06/16/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Ceres Main	Surface water	0.00	0.00	0.00	11.83	945,000.00 *gal*
Application event totals		0.00	0.00	0.00	11.83	

08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	110.52	37.06	179.13	743.54	108,000.00 *gal*
TID Ceres Main	Surface water	0.00	0.00	0.00	10.14	810,000.00 *gal*
Application event totals		110.52	37.06	179.13	753.68	

08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Ceres Main	Surface water	0.00	0.00	0.00	10.14	810,000.00 *gal*
Application event totals		0.00	0.00	0.00	10.14	

09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	110.52	37.06	179.13	743.54	108,000.00 *gal*
TID Ceres Main	Surface water	0.00	0.00	0.00	10.14	810,000.00 *gal*
Application event totals		110.52	37.06	179.13	753.68	

09/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Ceres Main	Surface water	0.00	0.00	0.00	10.14	810,000.00 *gal*
Application event totals		0.00	0.00	0.00	10.14	

Jakes - 10/24/2022: Oats, silage-soft dough

Field name: Jakes

Crop: Oats, silage-soft dough

Plant date: 10/24/2022

Jakes - 10/24/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/06/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	101.50	33.91	114.91	461.90	108,000.00 *gal*
TID Ceres Main	Surface water	1.39	0.00	0.00	0.00	810,000.00 *gal*
Application event totals		102.89	33.91	114.91	461.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	67.67	22.61	76.61	307.93	72,000.00 *gal*
Application event totals		67.67	22.61	76.61	307.93	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	92.10	30.89	149.27	619.62	90,000.00 *gal*
TID Ceres Main	Surface water	0.00	0.00	0.00	8.45	675,000.00 *gal*
Application event totals		92.10	30.89	149.27	628.07	

Jakes - 06/12/2023: Corn, silage

Field name: Jakes

Crop: Corn, silage

Plant date: 06/12/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	165.78	55.60	268.69	1,115.31	162,000.00 *gal*
TID Ceres Main	Surface water	0.00	0.00	0.00	15.21	1,215,000.00 *gal*
Application event totals		165.78	55.60	268.69	1,130.52	

Jakes - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Ceres Main	Surface water	0.00	0.00	0.00	13.52	1,080,000.00 *gal*
Application event totals		0.00	0.00	0.00	13.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	110.52	37.06	179.13	743.54	108,000.00 *gal*
TID Ceres Main	Surface water	0.00	0.00	0.00	10.14	810,000.00 *gal*
Application event totals		110.52	37.06	179.13	753.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	110.52	37.06	179.13	743.54	108,000.00 *gal*
TID Ceres Main	Surface water	0.00	0.00	0.00	10.14	810,000.00 *gal*
Application event totals		110.52	37.06	179.13	753.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Ceres Main	Surface water	0.00	0.00	0.00	13.52	1,080,000.00 *gal*
Application event totals		0.00	0.00	0.00	13.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Ceres Main	Surface water	0.00	0.00	0.00	13.52	1,080,000.00 *gal*
Application event totals		0.00	0.00	0.00	13.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Ceres Main	Surface water	0.00	0.00	0.00	11.83	945,000.00 *gal*
Application event totals		0.00	0.00	0.00	11.83	

Jakes - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Ceres Main	Surface water	0.00	0.00	0.00	10.14	810,000.00 *gal*
Application event totals		0.00	0.00	0.00	10.14	

Lander - Duane Almeida - 10/22/2020: Alfalfa, hay

Field name: Lander - Duane Almeida

Crop: Alfalfa, hay

Plant date: 10/22/2020

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/22/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	486.13	160.89	745.63	0.00	160.00 *ton*
Application event totals		486.13	160.89	745.63	0.00	

04/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Ground water	0.00	0.00	0.00	19.01	2,430,000.00 *gal*
Application event totals		0.00	0.00	0.00	19.01	

05/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Ground water	0.00	0.00	0.00	14.26	1,822,500.00 *gal*
Application event totals		0.00	0.00	0.00	14.26	

06/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Ground water	0.00	0.00	0.00	15.84	2,025,000.00 *gal*
Application event totals		0.00	0.00	0.00	15.84	

Lander - Duane Almeida - 10/22/2020: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Ground water	0.00	0.00	0.00	17.43	2,227,500.00 *gal*
Application event totals		0.00	0.00	0.00	17.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Ground water	0.00	0.00	0.00	16.63	2,126,250.00 *gal*
Application event totals		0.00	0.00	0.00	16.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Ground water	0.00	0.00	0.00	11.09	1,417,500.00 *gal*
Application event totals		0.00	0.00	0.00	11.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Ground water	0.00	0.00	0.00	15.05	1,923,750.00 *gal*
Application event totals		0.00	0.00	0.00	15.05	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Ground water	0.00	0.00	0.00	13.47	1,721,250.00 *gal*
Application event totals		0.00	0.00	0.00	13.47	

Scale - 10/27/2022: Oats, silage-soft dough

Field name: Scale

Crop: Oats, silage-soft dough

Plant date: 10/27/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Scale - 10/27/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/06/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	101.50	33.91	114.91	461.90	108,000.00 *gal*
TID Ceres Main	Surface water	1.39	0.00	0.00	0.00	810,000.00 *gal*
Application event totals		102.89	33.91	114.91	461.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	67.67	22.61	76.61	307.93	72,000.00 *gal*
Application event totals		67.67	22.61	76.61	307.93	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	110.52	37.06	179.13	743.54	108,000.00 *gal*
TID Ceres Main	Surface water	0.00	0.00	0.00	10.14	810,000.00 *gal*
Application event totals		110.52	37.06	179.13	753.68	

Scale - 06/16/2023: Sudangrass, silage

Field name: Scale

Crop: Sudangrass, silage

Plant date: 06/16/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Ceres Main	Surface water	0.00	0.00	0.00	15.21	1,215,000.00 *gal*
Application event totals		0.00	0.00	0.00	15.21	

Scale - 06/16/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	147.36	49.42	238.83	991.39	144,000.00 *gal*
TID Ceres Main	Surface water	0.00	0.00	0.00	13.52	1,080,000.00 *gal*
Application event totals		147.36	49.42	238.83	1,004.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Ceres Main	Surface water	0.00	0.00	0.00	10.14	810,000.00 *gal*
Application event totals		0.00	0.00	0.00	10.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Ceres Main	Surface water	0.00	0.00	0.00	10.14	810,000.00 *gal*
Application event totals		0.00	0.00	0.00	10.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	110.52	37.06	179.13	743.54	108,000.00 *gal*
TID Ceres Main	Surface water	0.00	0.00	0.00	10.14	810,000.00 *gal*
Application event totals		110.52	37.06	179.13	753.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Ceres Main	Surface water	0.00	0.00	0.00	10.14	810,000.00 *gal*
Application event totals		0.00	0.00	0.00	10.14	

Scale - 06/16/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	110.52	37.06	179.13	743.54	108,000.00 *gal*
TID Ceres Main	Surface water	0.00	0.00	0.00	10.14	810,000.00 *gal*
Application event totals		110.52	37.06	179.13	753.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Ceres Main	Surface water	0.00	0.00	0.00	10.14	810,000.00 *gal*
Application event totals		0.00	0.00	0.00	10.14	

Tony's - 11/14/2022: Oats, silage-soft dough

Field name: Tony's

Crop: Oats, silage-soft dough

Plant date: 11/14/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/14/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator Solids	Separator solids	203.47	45.86	111.75	1,175.62	500.00 *ton*
Application event totals		203.47	45.86	111.75	1,175.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5	Surface water	7.38	0.00	0.00	21.92	6,480,000.00 *gal*
Application event totals		7.38	0.00	0.00	21.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5	Surface water	0.00	0.00	0.00	306.92	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	306.92	

Tony's - 06/13/2023: Corn, silage

Field name: Tony's

Crop: Corn, silage

Plant date: 06/13/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/26/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator Solids	Separator solids	124.72	30.86	105.19	0.00	430.00 *ton*
Application event totals		124.72	30.86	105.19	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5	Surface water	0.00	0.00	0.00	287.73	6,075,000.00 *gal*
Application event totals		0.00	0.00	0.00	287.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
NH3	Solid commercial fertilizer	59.86	0.00	0.00	0.00	
Application event totals		59.86	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5	Surface water	0.00	0.00	0.00	249.37	5,265,000.00 *gal*
Application event totals		0.00	0.00	0.00	249.37	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
NH3	Solid commercial fertilizer	59.86	0.00	0.00	0.00	
Application event totals		59.86	0.00	0.00	0.00	

Tony's - 06/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5	Surface water	0.00	0.00	0.00	230.19	4,860,000.00 *gal*
Application event totals		0.00	0.00	0.00	230.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
NH3	Solid commercial fertilizer	59.86	0.00	0.00	0.00	
Application event totals		59.86	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5	Surface water	0.00	0.00	0.00	239.78	5,062,500.00 *gal*
Application event totals		0.00	0.00	0.00	239.78	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5	Surface water	0.00	0.00	0.00	230.19	4,860,000.00 *gal*
Application event totals		0.00	0.00	0.00	230.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5	Surface water	0.00	0.00	0.00	230.19	4,860,000.00 *gal*
Application event totals		0.00	0.00	0.00	230.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5	Surface water	0.00	0.00	0.00	211.00	4,455,000.00 *gal*
Application event totals		0.00	0.00	0.00	211.00	

Tony's - 06/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5	Surface water	0.00	0.00	0.00	191.82	4,050,000.00 *gal*
Application event totals		0.00	0.00	0.00	191.82	

Triangle - 11/14/2022: Oats, silage-soft dough

Field name: Triangle

Crop: Oats, silage-soft dough

Plant date: 11/14/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/19/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	105.26	35.17	119.17	479.00	252,000.00 *gal*
TID Ceres Main	Surface water	1.44	0.00	0.00	0.00	1,890,000.00 *gal*
Application event totals		106.70	35.17	119.17	479.00	

04/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	98.24	32.95	159.22	660.92	216,000.00 *gal*
TID Ceres Main	Surface water	0.00	0.00	0.00	9.01	1,620,000.00 *gal*
Application event totals		98.24	32.95	159.22	669.94	

Triangle - 06/16/2023: Sudangrass, silage

Field name: Triangle

Crop: Sudangrass, silage

Plant date: 06/16/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Triangle - 06/16/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Ceres Main	Surface water	0.00	0.00	0.00	15.02	2,700,000.00 *gal*
Application event totals		0.00	0.00	0.00	15.02	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	130.98	43.93	212.30	881.23	288,000.00 *gal*
TID Ceres Main	Surface water	0.00	0.00	0.00	12.02	2,160,000.00 *gal*
Application event totals		130.98	43.93	212.30	893.25	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Ceres Main	Surface water	0.00	0.00	0.00	12.02	2,160,000.00 *gal*
Application event totals		0.00	0.00	0.00	12.02	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Ceres Main	Surface water	0.00	0.00	0.00	10.51	1,890,000.00 *gal*
Application event totals		0.00	0.00	0.00	10.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	114.61	38.44	185.76	771.08	252,000.00 *gal*
TID Ceres Main	Surface water	0.00	0.00	0.00	10.51	1,890,000.00 *gal*
Application event totals		114.61	38.44	185.76	781.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Ceres Main	Surface water	0.00	0.00	0.00	9.76	1,755,000.00 *gal*
Application event totals		0.00	0.00	0.00	9.76	

Triangle - 06/16/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	98.24	32.95	159.22	660.92	216,000.00 *gal*
TID Ceres Main	Surface water	0.00	0.00	0.00	9.01	1,620,000.00 *gal*
Application event totals		98.24	32.95	159.22	669.94	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/25/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Ceres Main	Surface water	0.00	0.00	0.00	9.01	1,620,000.00 *gal*
Application event totals		0.00	0.00	0.00	9.01	

Verduga - 10/22/2020: Alfalfa, hay

Field name: Verduga

Crop: Alfalfa, hay

Plant date: 10/22/2020

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/22/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	486.13	160.89	745.63	0.00	160.00 *ton*
Application event totals		486.13	160.89	745.63	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Turlock Main	Surface water	0.00	0.00	0.00	17.85	2,632,500.00 *gal*
Application event totals		0.00	0.00	0.00	17.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Turlock Main	Surface water	0.00	0.00	0.00	16.48	2,430,000.00 *gal*
Application event totals		0.00	0.00	0.00	16.48	

Verduga - 10/22/2020: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Turlock Main	Surface water	0.00	0.00	0.00	16.48	2,430,000.00 *gal*
Application event totals		0.00	0.00	0.00	16.48	

07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Turlock Main	Surface water	0.00	0.00	0.00	16.48	2,430,000.00 *gal*
Application event totals		0.00	0.00	0.00	16.48	

07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Turlock Main	Surface water	0.00	0.00	0.00	21.97	3,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	21.97	

08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Turlock Main	Surface water	0.00	0.00	0.00	18.31	2,700,000.00 *gal*
Application event totals		0.00	0.00	0.00	18.31	

09/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Turlock Main	Surface water	0.00	0.00	0.00	20.14	2,970,000.00 *gal*
Application event totals		0.00	0.00	0.00	20.14	

10/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Turlock Main	Surface water	0.00	0.00	0.00	18.31	2,700,000.00 *gal*
Application event totals		0.00	0.00	0.00	18.31	

Washington - Sandoval - 01/28/2016: Almond, in shell

Washington - Sandoval - 01/28/2016: Almond, in shell

Field name: Washington - Sandoval

Crop: Almond, in shell

Plant date: 01/28/2016

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/12/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	48.00	0.00	0.00	0.00	
Application event totals		48.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5	Surface water	0.00	0.00	0.00	138.01	1,417,500.00 *gal*
Application event totals		0.00	0.00	0.00	138.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5	Surface water	0.00	0.00	0.00	226.72	2,328,750.00 *gal*
Application event totals		0.00	0.00	0.00	226.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5	Surface water	0.00	0.00	0.00	197.15	2,025,000.00 *gal*
Application event totals		0.00	0.00	0.00	197.15	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/13/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	48.00	0.00	0.00	0.00	
Application event totals		48.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5	Surface water	0.00	0.00	0.00	216.87	2,227,500.00 *gal*
Application event totals		0.00	0.00	0.00	216.87	

Washington - Sandoval - 01/28/2016: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5	Surface water	0.00	0.00	0.00	341.73	3,510,000.00 *gal*
Application event totals		0.00	0.00	0.00	341.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5	Surface water	0.00	0.00	0.00	381.16	3,915,000.00 *gal*
Application event totals		0.00	0.00	0.00	381.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5	Surface water	0.00	0.00	0.00	276.01	2,835,000.00 *gal*
Application event totals		0.00	0.00	0.00	276.01	

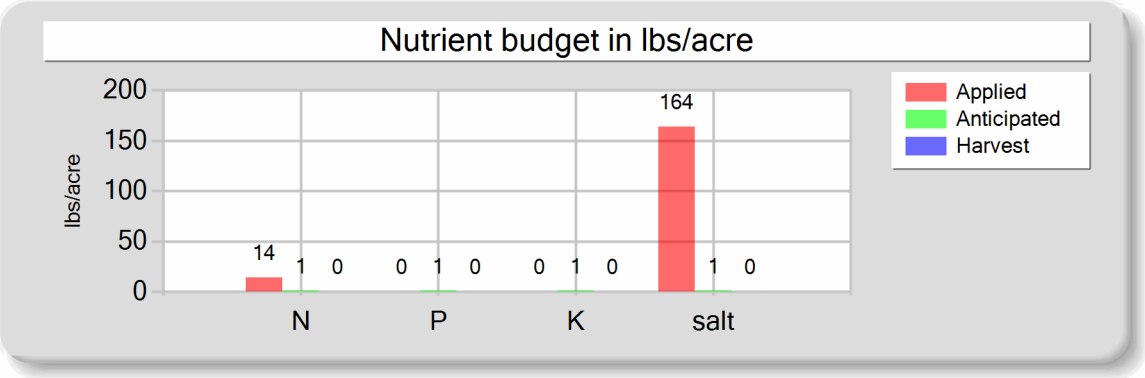
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5	Surface water	0.00	0.00	0.00	315.44	3,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	315.44	

B. NUTRIENT BUDGET

Dad's - 01/28/2021: Almond, in shell

Field name: Dad's Crop: Almond, in shell Plant date: 01/28/2021



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	164.03
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	14.00	0.00	0.00	164.03
Anticipated crop nutrient removal	1.00	1.00	1.00	1.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	14.00	0.00	0.00	164.03
Applied to removed ratio	0.00	0.00	0.00	0.00

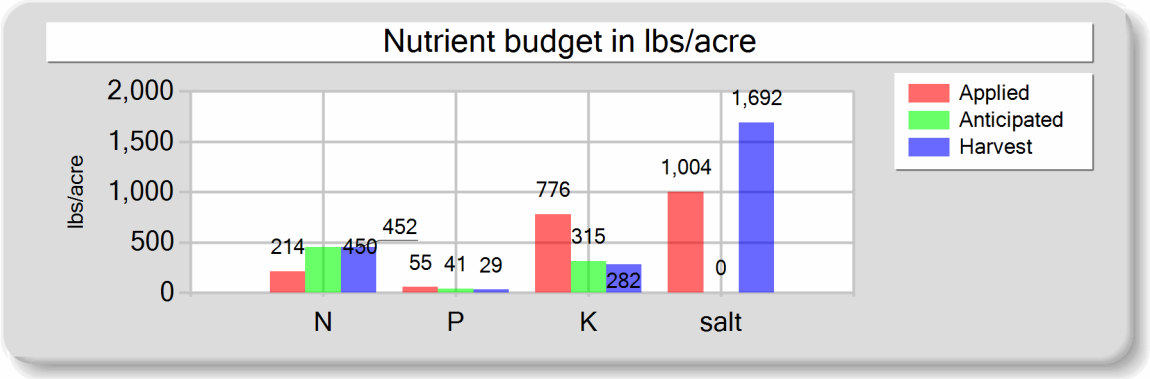
Fresh water applied	
36,288,000.00	*gallons*
1,336.36	*acre-inches*
27.84	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Hirdes East - 10/20/2020: Alfalfa, hay

Field name: Hirdes East | Crop: Alfalfa, hay | Plant date: 10/20/2020



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	600.30	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	199.71	54.51	175.89	723.72
Fresh water	0.00	0.00	0.00	280.22
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	213.71	54.51	776.19	1,003.93
Anticipated crop nutrient removal	450.00	40.50	315.00	0.00
Actual crop nutrient removal	452.41	29.14	281.75	1,691.56
Nutrient balance	-238.70	25.37	494.44	-687.63
Applied to removed ratio	0.47	1.87	2.75	0.59

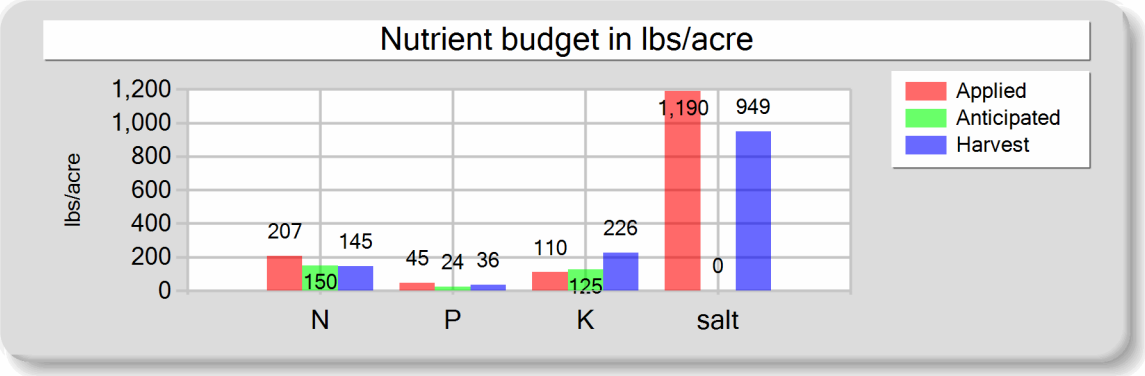
Fresh water applied	
116,235,000.00	*gallons*
4,280.54	*acre-inches*
47.56	*inches/acre*

Process wastewater applied	
988,000.00	*gallons*
36.38	*acre-inches*
0.40	*inches/acre*

Total harvests for the crop	
1	*harvests*

Hirdes West - 11/22/2022: Oats, silage-soft dough

Field name: Hirdes West | Crop: Oats, silage-soft dough | Plant date: 11/22/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	200.23	45.13	109.97	1,156.86
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	32.72
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	207.23	45.13	109.97	1,189.58
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	144.63	36.16	225.98	949.11
Nutrient balance	62.60	8.97	-116.01	240.47
Applied to removed ratio	1.43	1.25	0.49	1.25

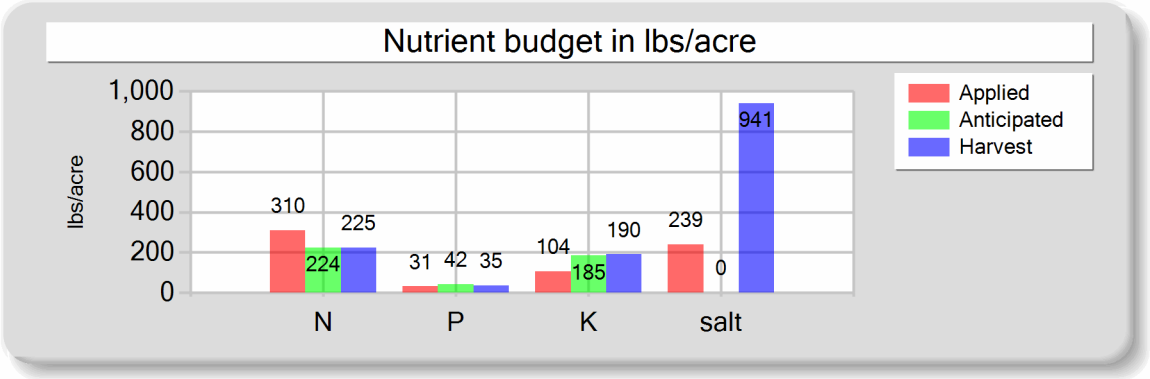
Fresh water applied	
55,890,000.00	*gallons*
2,058.24	*acre-inches*
10.95	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Hirdes West - 06/12/2023: Corn, silage

Field name: Hirdes West | Crop: Corn, silage | Plant date: 06/12/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	179.58	0.00	0.00	0.00
Dry manure	123.87	30.65	104.48	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	238.69
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	310.45	30.65	104.48	238.69
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	224.79	34.58	190.21	940.65
Nutrient balance	85.66	-3.93	-85.73	-701.96
Applied to removed ratio	1.38	0.89	0.55	0.25

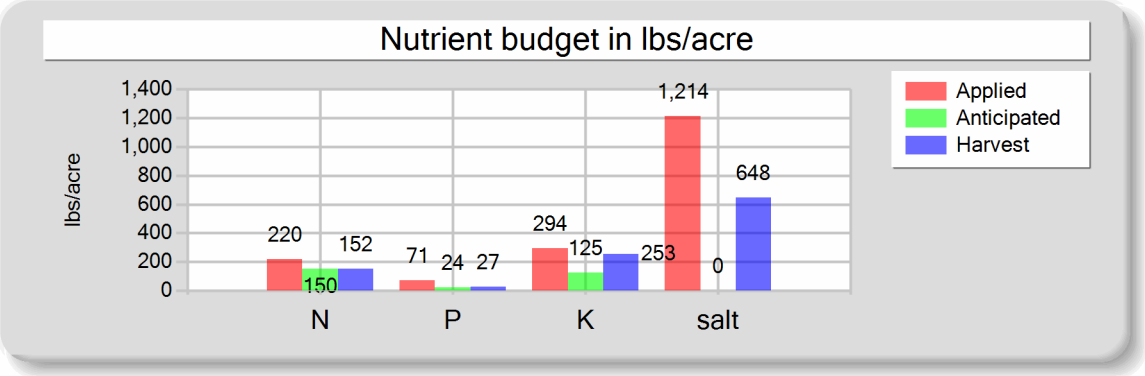
Fresh water applied	
206,820,000.00	*gallons*
7,616.48	*acre-inches*
40.51	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Home - 10/27/2022: Oats, silage-soft dough

Field name: Home | Crop: Oats, silage-soft dough | Plant date: 10/27/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	212.02	70.97	294.04	1,205.44
Fresh water	1.39	0.00	0.00	8.45
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	220.41	70.97	294.04	1,213.88
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	152.23	27.01	252.89	648.19
Nutrient balance	68.18	43.97	41.15	565.70
Applied to removed ratio	1.45	2.63	1.16	1.87

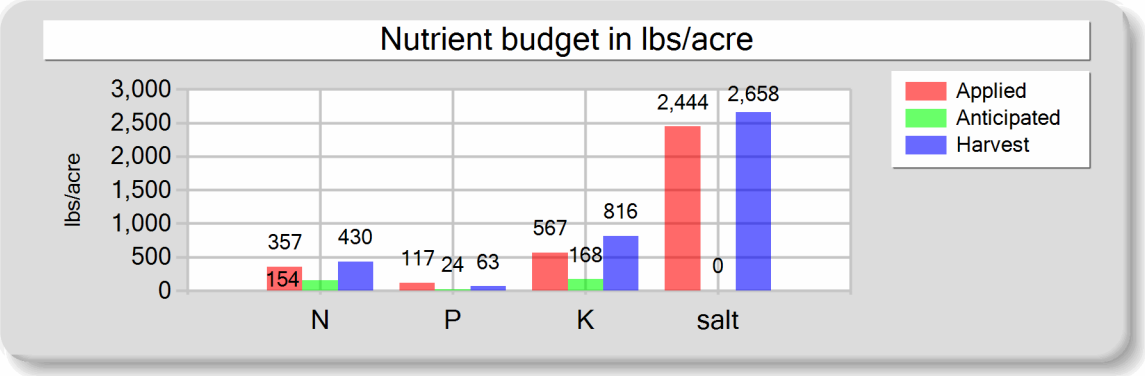
Fresh water applied	
1,485,000.00	*gallons*
54.69	*acre-inches*
6.84	*inches/acre*

Process wastewater applied	
216,000.00	*gallons*
7.95	*acre-inches*
0.99	*inches/acre*

Total harvests for the crop	
1	*harvests*

Home - 06/16/2023: Sudangrass, silage

Field name: Home | Crop: Sudangrass, silage | Plant date: 06/16/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	349.97	117.37	567.23	2,354.54
Fresh water	0.00	0.00	0.00	89.56
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	356.97	117.37	567.23	2,444.10
Anticipated crop nutrient removal	154.00	23.80	168.00	0.00
Actual crop nutrient removal	430.07	63.03	815.65	2,657.54
Nutrient balance	-73.10	54.34	-248.42	-213.43
Applied to removed ratio	0.83	1.86	0.70	0.92

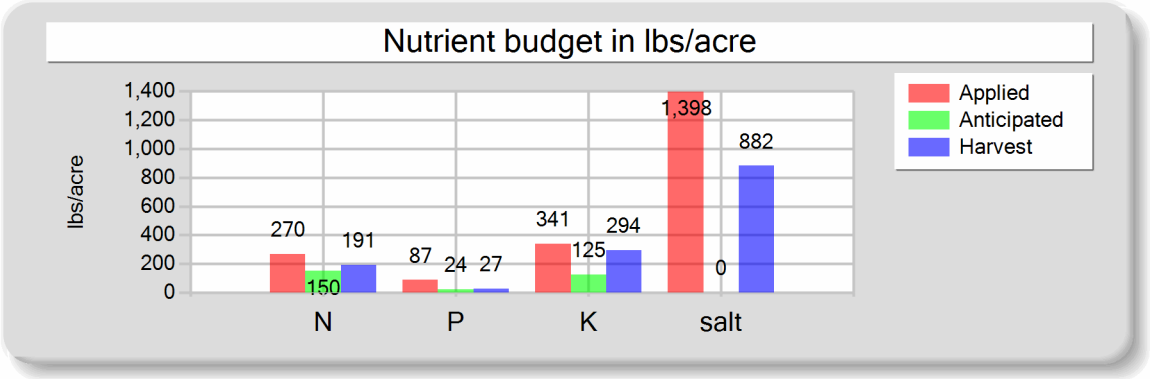
Fresh water applied	
7,155,000.00	*gallons*
263.49	*acre-inches*
32.94	*inches/acre*

Process wastewater applied	
342,000.00	*gallons*
12.59	*acre-inches*
1.57	*inches/acre*

Total harvests for the crop	
1	*harvests*

Jakes - 10/24/2022: Oats, silage-soft dough

Field name: Jakes | Crop: Oats, silage-soft dough | Plant date: 10/24/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	261.27	87.40	340.79	1,389.44
Fresh water	1.39	0.00	0.00	8.45
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	269.66	87.40	340.79	1,397.89
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	191.10	27.30	294.00	882.00
Nutrient balance	78.56	60.10	46.79	515.89
Applied to removed ratio	1.41	3.20	1.16	1.58

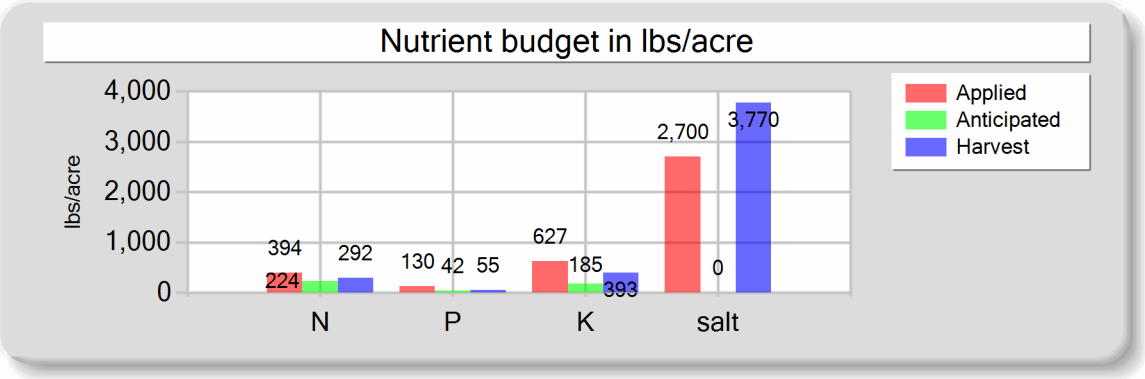
Fresh water applied	
1,485,000.00	*gallons*
54.69	*acre-inches*
6.84	*inches/acre*

Process wastewater applied	
270,000.00	*gallons*
9.94	*acre-inches*
1.24	*inches/acre*

Total harvests for the crop	
1	*harvests*

Jakes - 06/12/2023: Corn, silage

Field name: Jakes Crop: Corn, silage Plant date: 06/12/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	386.81	129.73	626.94	2,602.39
Fresh water	0.00	0.00	0.00	98.01
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	393.81	129.73	626.94	2,700.40
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	291.53	54.93	392.93	3,770.05
Nutrient balance	102.28	74.80	234.01	-1,069.65
Applied to removed ratio	1.35	2.36	1.60	0.72

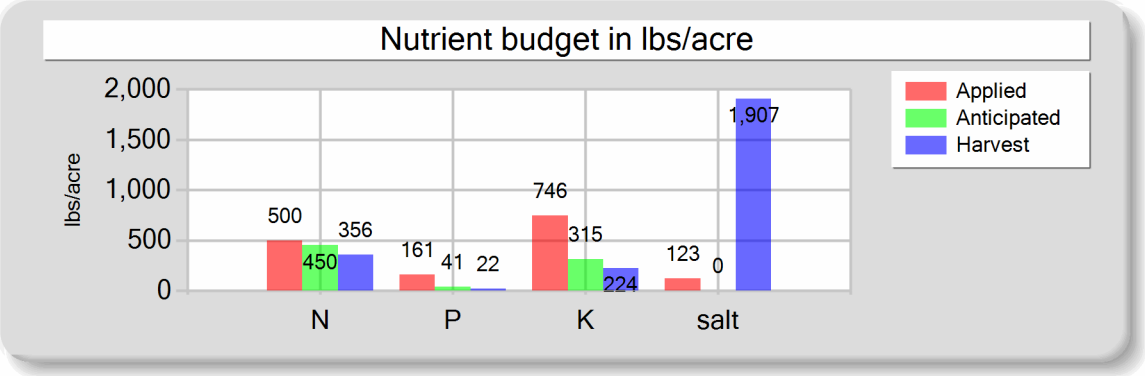
Fresh water applied	
7,830,000.00	*gallons*
288.35	*acre-inches*
36.04	*inches/acre*

Process wastewater applied	
378,000.00	*gallons*
13.92	*acre-inches*
1.74	*inches/acre*

Total harvests for the crop	
1	*harvests*

Lander - Duane Almeida - 10/22/2020: Alfalfa, hay

Field name: Lander - Duane Almeida Crop: Alfalfa, hay Plant date: 10/22/2020



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	486.13	160.89	745.63	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	122.78
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	500.13	160.89	745.63	122.78
Anticipated crop nutrient removal	450.00	40.50	315.00	0.00
Actual crop nutrient removal	356.16	22.43	224.30	1,906.86
Nutrient balance	143.97	138.46	521.33	-1,784.08
Applied to removed ratio	1.40	7.17	3.32	0.06

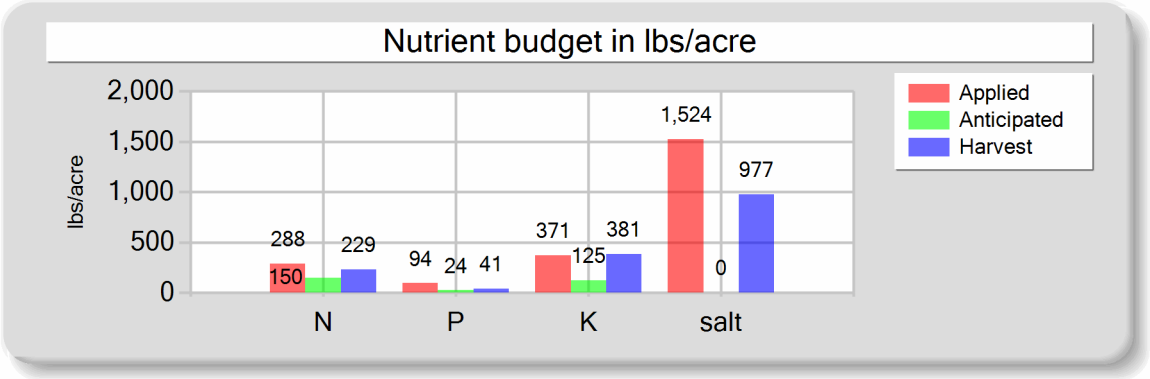
Fresh water applied	
15,693,750.00	*gallons*
577.95	*acre-inches*
36.12	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Scale - 10/27/2022: Oats, silage-soft dough

Field name: Scale | Crop: Oats, silage-soft dough | Plant date: 10/27/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	279.69	93.58	370.64	1,513.37
Fresh water	1.39	0.00	0.00	10.14
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	288.08	93.58	370.64	1,523.50
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	229.40	40.70	381.10	976.80
Nutrient balance	58.68	52.88	-10.46	546.70
Applied to removed ratio	1.26	2.30	0.97	1.56

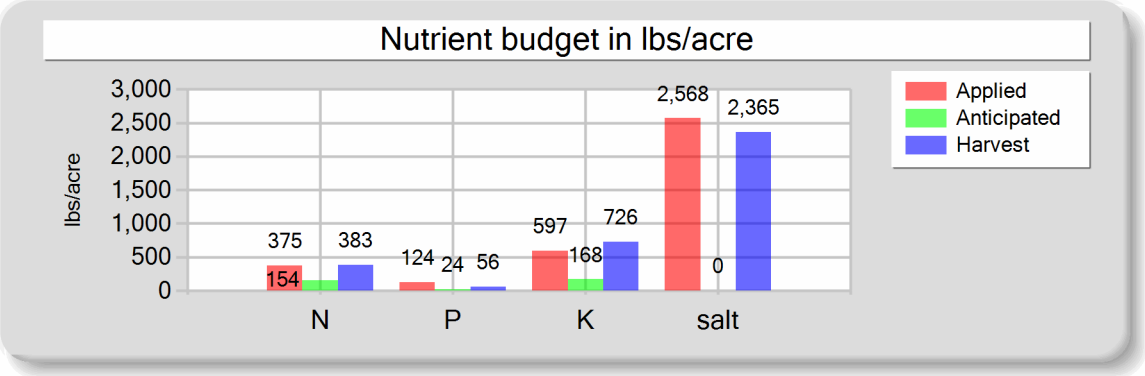
Fresh water applied	
1,620,000.00	*gallons*
59.66	*acre-inches*
7.46	*inches/acre*

Process wastewater applied	
288,000.00	*gallons*
10.61	*acre-inches*
1.33	*inches/acre*

Total harvests for the crop	
1	*harvests*

Scale - 06/16/2023: Sudangrass, silage

Field name: Scale | Crop: Sudangrass, silage | Plant date: 06/16/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	368.39	123.55	597.08	2,478.47
Fresh water	0.00	0.00	0.00	89.56
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	375.39	123.55	597.08	2,568.03
Anticipated crop nutrient removal	154.00	23.80	168.00	0.00
Actual crop nutrient removal	382.68	56.08	725.78	2,364.72
Nutrient balance	-7.29	67.46	-128.70	203.30
Applied to removed ratio	0.98	2.20	0.82	1.09

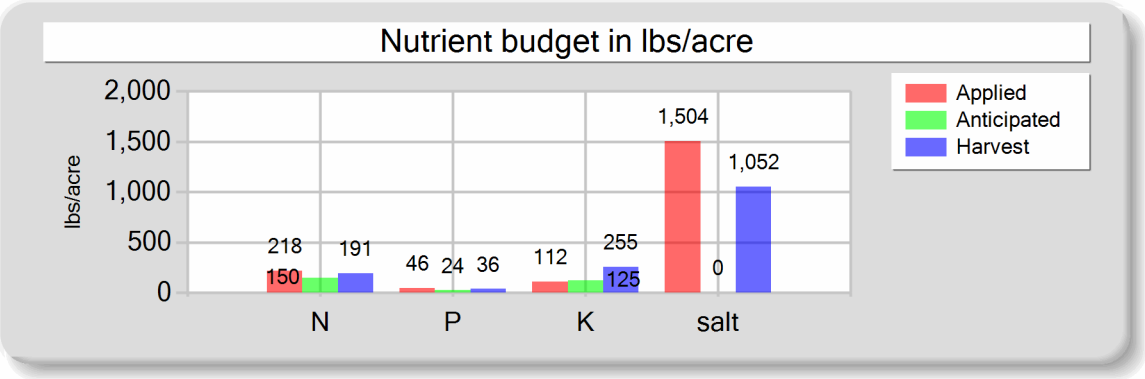
Fresh water applied	
7,155,000.00	*gallons*
263.49	*acre-inches*
32.94	*inches/acre*

Process wastewater applied	
360,000.00	*gallons*
13.26	*acre-inches*
1.66	*inches/acre*

Total harvests for the crop	
1	*harvests*

Tony's - 11/14/2022: Oats, silage-soft dough

Field name: Tony's Crop: Oats, silage-soft dough Plant date: 11/14/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	203.47	45.86	111.75	1,175.62
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	7.38	0.00	0.00	328.84
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	217.85	45.86	111.75	1,504.46
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	191.36	36.45	255.15	1,052.50
Nutrient balance	26.49	9.41	-143.40	451.96
Applied to removed ratio	1.14	1.26	0.44	1.43

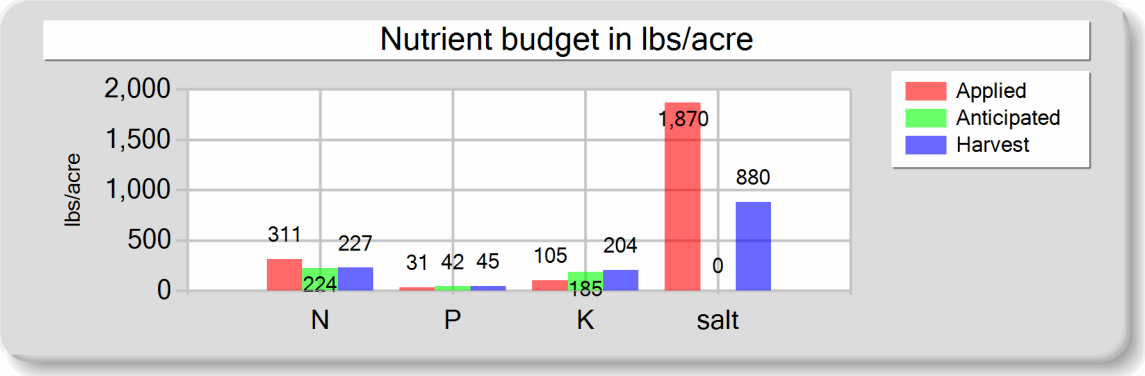
Fresh water applied	
12,960,000.00	*gallons*
477.27	*acre-inches*
12.90	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Tony's - 06/13/2023: Corn, silage

Field name: Tony's Crop: Corn, silage Plant date: 06/13/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	179.58	0.00	0.00	0.00
Dry manure	124.72	30.86	105.19	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,870.27
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	311.30	30.86	105.19	1,870.27
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	226.87	45.37	204.18	879.80
Nutrient balance	84.43	-14.51	-98.99	990.47
Applied to removed ratio	1.37	0.68	0.52	2.13

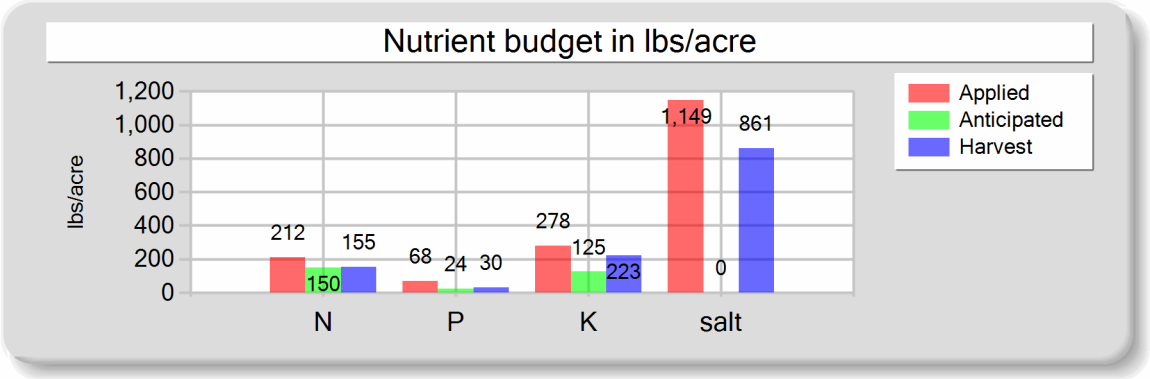
Fresh water applied	
39,487,500.00	*gallons*
1,454.19	*acre-inches*
39.30	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Triangle - 11/14/2022: Oats, silage-soft dough

Field name: Triangle | Crop: Oats, silage-soft dough | Plant date: 11/14/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	203.50	68.11	278.39	1,139.93
Fresh water	1.44	0.00	0.00	9.01
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	211.94	68.11	278.39	1,148.94
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	154.89	30.22	222.89	861.33
Nutrient balance	57.05	37.89	55.50	287.61
Applied to removed ratio	1.37	2.25	1.25	1.33

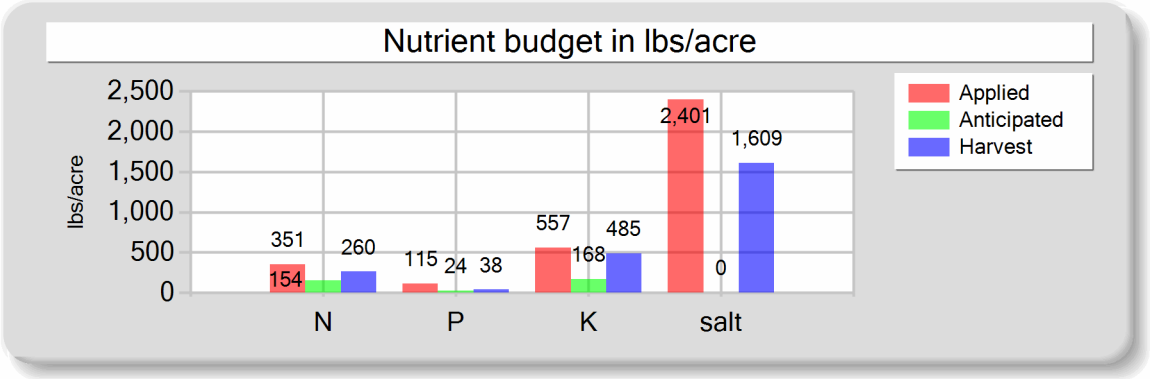
Fresh water applied	
3,510,000.00	*gallons*
129.26	*acre-inches*
7.18	*inches/acre*

Process wastewater applied	
468,000.00	*gallons*
17.23	*acre-inches*
0.96	*inches/acre*

Total harvests for the crop	
1	*harvests*

Triangle - 06/16/2023: Sudangrass, silage

Field name: Triangle | Crop: Sudangrass, silage | Plant date: 06/16/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	343.83	115.31	557.28	2,313.23
Fresh water	0.00	0.00	0.00	87.87
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	350.83	115.31	557.28	2,401.11
Anticipated crop nutrient removal	154.00	23.80	168.00	0.00
Actual crop nutrient removal	260.27	37.96	485.29	1,608.99
Nutrient balance	90.56	77.36	71.99	792.12
Applied to removed ratio	1.35	3.04	1.15	1.49

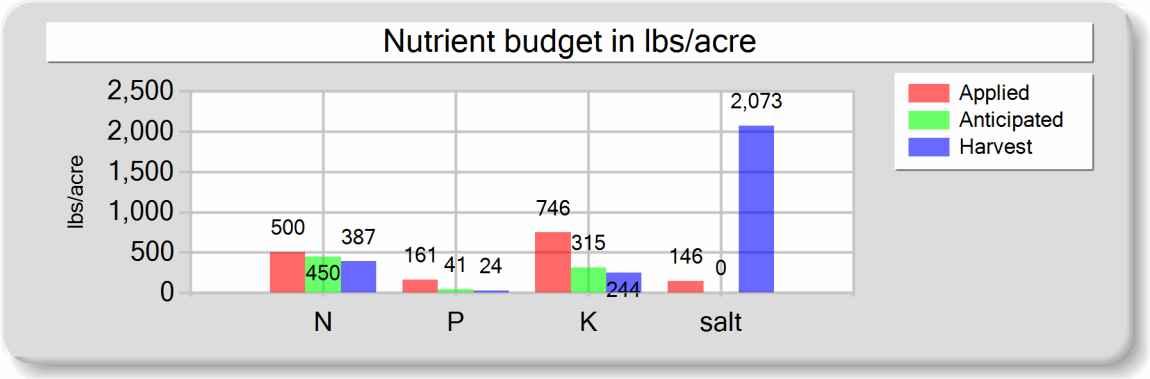
Fresh water applied	
15,795,000.00	*gallons*
581.68	*acre-inches*
32.32	*inches/acre*

Process wastewater applied	
756,000.00	*gallons*
27.84	*acre-inches*
1.55	*inches/acre*

Total harvests for the crop	
1	*harvests*

Verduga - 10/22/2020: Alfalfa, hay

Field name: Verduga Crop: Alfalfa, hay Plant date: 10/22/2020



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	486.13	160.89	745.63	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	146.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	500.13	160.89	745.63	146.00
Anticipated crop nutrient removal	450.00	40.50	315.00	0.00
Actual crop nutrient removal	387.27	24.39	243.89	2,073.44
Nutrient balance	112.86	136.50	501.74	-1,927.44
Applied to removed ratio	1.29	6.60	3.06	0.07

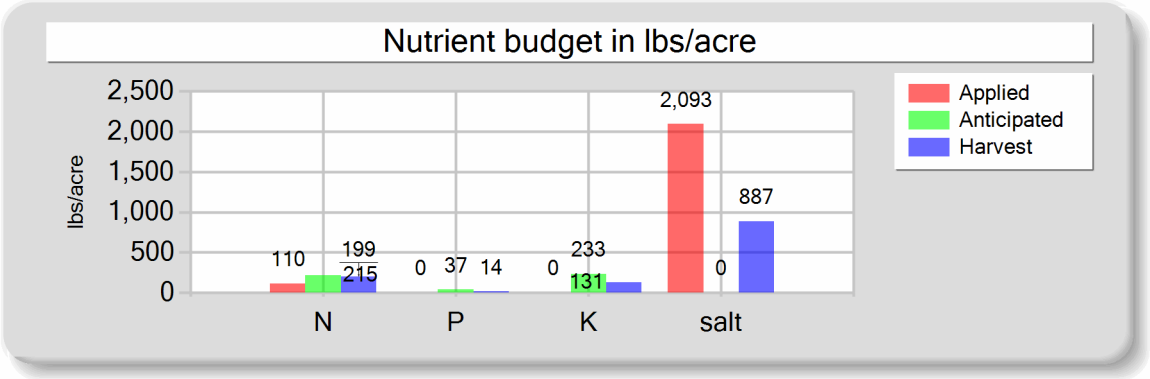
Fresh water applied	
21,532,500.00	*gallons*
792.97	*acre-inches*
49.56	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Washington - Sandoval - 01/28/2016: Almond, in shell

Field name: Washington - Sandoval Crop: Almond, in shell Plant date: 01/28/2016



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	96.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	2,093.08
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	110.00	0.00	0.00	2,093.08
Anticipated crop nutrient removal	215.00	36.50	232.50	0.00
Actual crop nutrient removal	198.98	13.98	130.73	886.96
Nutrient balance	-88.98	-13.98	-130.73	1,206.12
Applied to removed ratio	0.55	0.00	0.00	2.36

Fresh water applied	
21,498,750.00	*gallons*
791.73	*acre-inches*
43.98	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Corral Solids

Sample and source description: Corral Solids

Sample date: 10/13/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 13.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	28,100.00	9,300.00	43,100.00							0.00
DL	100.00	100.00	100.00							0.01

Separator Solids

Sample and source description: Separator Solids

Sample date: 10/13/2022 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 76.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	31,500.00	7,100.00	17,300.00	18,000.00	7,100.00	8,700.00	5,900.00	8,000.00		18.20
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 05/18/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 10.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,100.00	6,100.00	18,600.00							0.00
DL	100.00	100.00	100.00							0.01

Separator Solids

Sample and source description: Separator Solids

Sample date: 05/18/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,800.00	4,900.00	16,700.00							0.00
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 10/19/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 44.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,600.00	8,400.00	6,700.00							0.00
DL	100.00	100.00	100.00							0.01

Separator Solids

Sample and source description: Separator Solids

Sample date: 10/19/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 78.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	27,000.00	8,100.00	11,700.00							0.00
DL	100.00	100.00	100.00							0.01

B. PROCESS WASTEWATER ANALYSES

WWQ3

Sample and source description: WWQ3

Sample date: 08/18/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.70

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	981.00	606.00	0.00	0.00	329.00	1,590.00	600.00	326.00	970.00	12,800.00	0.00	22.60	1,390.00	14,700.00	6,600
DL	1.00	0.50	0.50	0.50	0.10	0.50	0.10	0.10	1.00	5.00	2.00	0.50	0.20	10.00	10

WWQ4

Sample and source description: WWQ4

Sample date: 10/13/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.50

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	901.00	467.00	0.00	0.00	301.00	1,020.00								11,400.00	4,100
DL	1.00	0.50	0.50	0.50	0.10	0.50								10.00	10

WWQ1

Sample and source description: WWQ1

Sample date: 02/23/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.50

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	2,180.00	823.00	0.00	0.00	595.00	1,920.00								16,200.00	7,900
DL	1.00	0.50	0.50	0.50	0.10	0.50								10.00	10

WWQ2

Sample and source description: WWQ2

Sample date: 05/18/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.70

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	2,000.00	878.00	0.00	0.00	550.00	1,780.00	1,110.00	486.00	1,090.00	10,000.00	0.00	6.00	1,400.00	19,300.00	7,900
DL	1.00	0.50	0.50	0.50	0.10	0.50	0.10	0.10	1.00	5.00	2.00	0.50	0.20	10.00	10

WWQ4

Sample and source description: WWQ4

Sample date: 11/09/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.50

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,060.00	548.00	0.00	0.00	215.00	1,130.00								12,600.00	6,450
DL	1.00	0.50	0.50	0.50	0.10	0.50								10.00	10

C. FRESH WATER ANALYSES

TID Ceres Main

TID Ceres Main

Sample description: TID Ceres Main

Sample date: 07/22/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.64	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

TID Ceres Main

Sample description: TID Ceres Main

Sample date: 07/25/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								30.30	12
DL	1.00	0.50	0.50								10.00	10

TID Lateral 4

TID Lateral 4

TID Lateral 4

Sample description: TID Lateral 4

Sample date: 07/25/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								31.10	26
DL	1.00	0.50	0.50								10.00	10

TID Lateral 5

TID Lateral 5

Sample description: TID Lateral 5

Sample date: 08/23/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	5.05	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	613.00	15
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

TID Lateral 6

TID Lateral 6

Sample description: TID Lateral 6

Sample date: 08/23/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1,060.00	15
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

TID Lateral 6

TID Lateral 6

Sample description: TID Lateral 6

Sample date: 08/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	210
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

TID Lateral 6

Sample description: TID Lateral 6

Sample date: 08/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	215
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

TID Turlock Main

TID Turlock Main

Sample description: TID Turlock Main

Sample date: 07/22/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

TID Turlock Main

TID Turlock Main

Sample description: TID Turlock Main

Sample date: 07/25/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								30.30	13
DL	1.00	0.50	0.50								10.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Dad's - 01/28/2021: Almond, in shell

Non Bearing ALmonds

Sample and source description: Non Bearing ALmonds

Sample date: 12/31/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 0.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	0.00	0.00	0.00		0.00
DL	100.00	100.00	100.00		0.01

Hirdes East - 10/20/2020: Alfalfa, hay

Hirdes East - 10/20/2020: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 11/21/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 10.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	28,100.00	1,810.00	17,500.00		11.70
DL	100.00	100.00	100.00		0.01

Hirdes West - 11/22/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/24/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,200.00	800.00	5,000.00		7.00
DL	100.00	100.00	100.00		0.01

Forage

Sample and source description: Forage

Sample date: 05/25/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	2,400.00	600.00	4,800.00		7.00
DL	100.00	100.00	100.00		0.01

Hirdes West - 06/12/2023: Corn, silage

Hirdes West - 06/12/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage
Sample date: 10/12/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 68.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,900.00	600.00	3,300.00		5.10
DL	100.00	100.00	100.00		0.01

Corn Silage

Sample and source description: Corn Silage
Sample date: 10/12/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 68.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,700.00	700.00	3,600.00		5.10
DL	100.00	100.00	100.00		0.01

Home - 10/27/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage
Sample date: 04/17/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,200.00	1,100.00	10,300.00		8.80
DL	100.00	100.00	100.00		0.01

Home - 06/16/2023: Sudangrass, silage

Home - 06/16/2023: Sudangrass, silage

Sudan

Sample and source description: Sudan

Sample date: 10/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 55.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,600.00	1,700.00	22,000.00		16.00
DL	100.00	100.00	100.00		0.01

Jakes - 10/24/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 04/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,100.00	1,300.00	14,000.00		14.00
DL	100.00	100.00	100.00		0.01

Jakes - 06/12/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 57.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,900.00	1,300.00	9,300.00		20.80
DL	100.00	100.00	100.00		0.01

Lander - Duane Almeida - 10/22/2020: Alfalfa, hay

ALfalfa

Sample and source description: ALfalfa

Sample date: 11/21/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 9.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,200.00	1,650.00	16,500.00		15.50
DL	100.00	100.00	100.00		0.01

Scale - 10/27/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 04/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,200.00	1,100.00	10,300.00		8.80
DL	100.00	100.00	100.00		0.01

Tony's - 11/14/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/16/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,200.00	800.00	5,600.00		7.70
DL	100.00	100.00	100.00		0.01

Tony's - 06/13/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 72.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,000.00	800.00	3,600.00		5.60
DL	100.00	100.00	100.00		0.01

Triangle - 11/14/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/16/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,100.00	800.00	5,900.00		7.60
DL	100.00	100.00	100.00		0.01

Triangle - 06/16/2023: Sudangrass, silage

Sudan

Sample and source description: Sudan

Sample date: 10/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 59.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,600.00	1,400.00	17,900.00		14.80
DL	100.00	100.00	100.00		0.01

Verduga - 10/22/2020: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 11/21/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 9.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,200.00	1,650.00	16,500.00		15.50
DL	100.00	100.00	100.00		0.01

Washington - Sandoval - 01/28/2016: Almond, in shell

Almonds

Sample and source description: Almonds

Sample date: 10/31/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 7.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,200.00	1,700.00	15,900.00		11.70
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

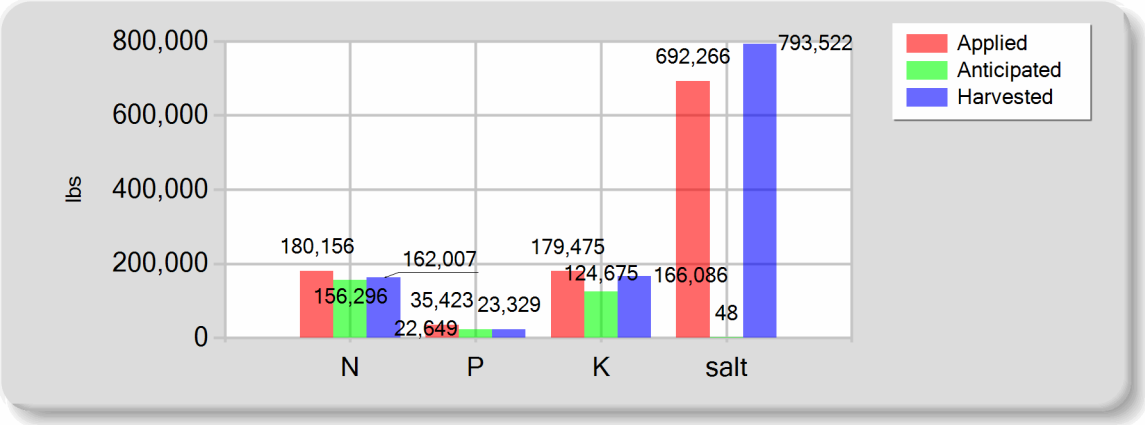
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

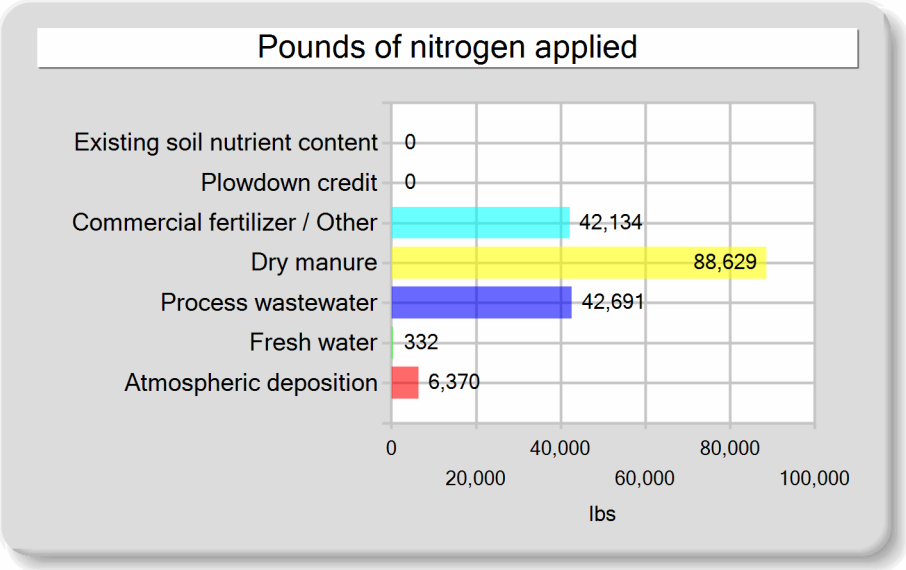
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

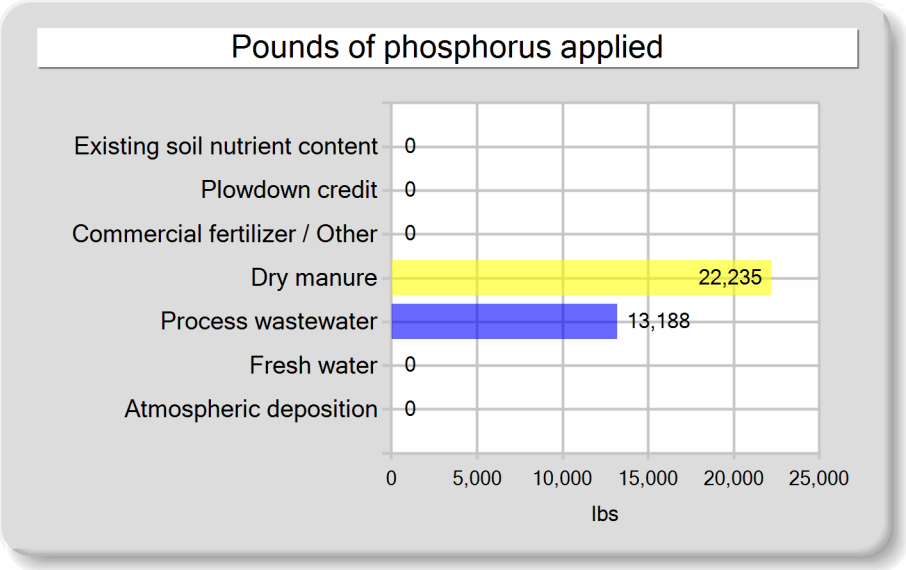
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	42,133.50	0.00	54,027.00	0.00
Dry manure	88,629.32	22,234.96	72,202.00	260,988.00
Process wastewater	42,691.00	13,188.12	53,245.94	219,640.40
Fresh water	332.20	0.00	0.00	211,637.87
Atmospheric deposition	6,370.00	0.00	0.00	0.00
Total nutrients applied	180,156.02	35,423.08	179,474.94	692,266.27
Anticipated crop nutrient removal	156,296.00	22,649.20	124,674.90	48.00
Actual crop nutrient removal	162,006.73	23,329.44	166,085.70	793,521.79
Nutrient balance	18,149.29	12,093.64	13,389.24	-101,255.52
Applied to removed ratio	1.11	1.52	1.08	0.87

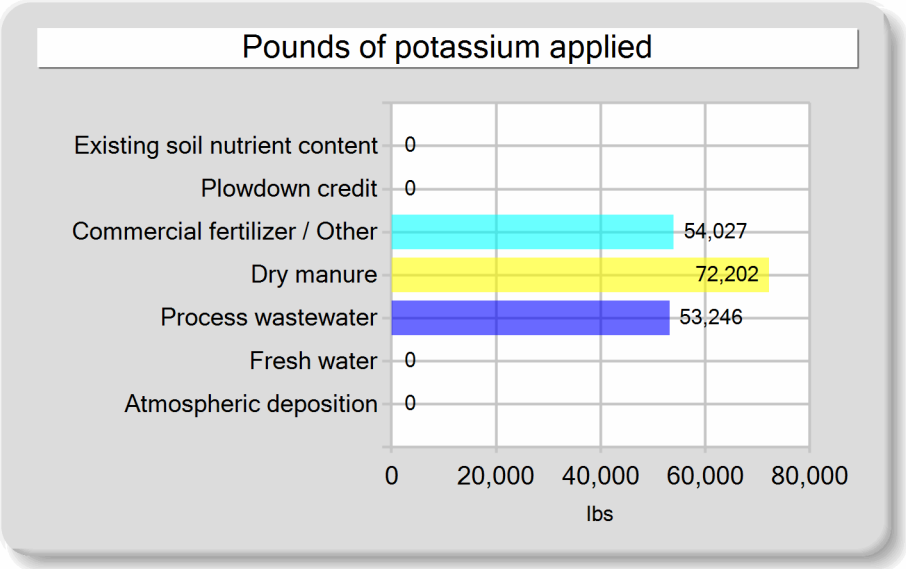
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

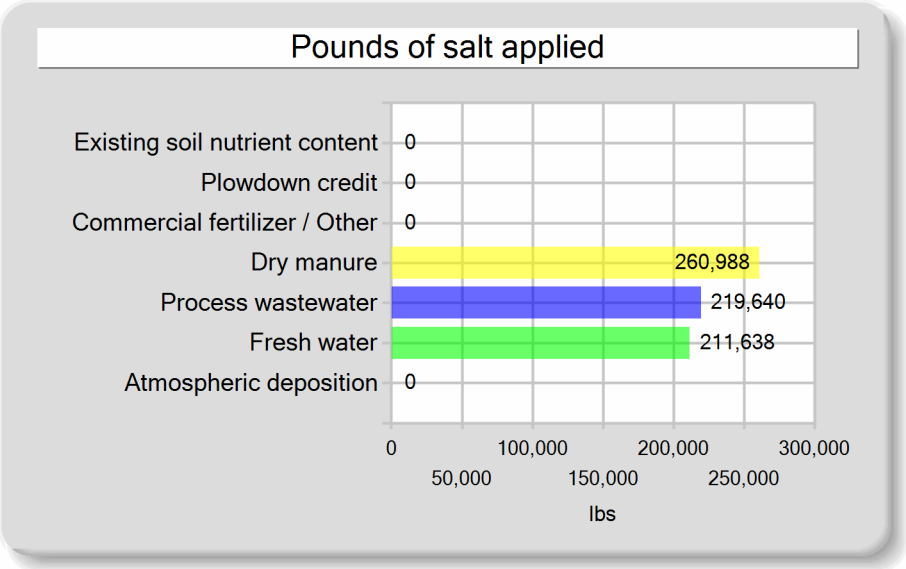


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Jim Vieira	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
4-18-2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

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LABORATORY INC

GJC Dairy
P.O. Box 2397
Turlock, CA 95381

Account# 00-0025382
Account Manager: Ben Nydam
Submitted By: F & R Ag Services
Ranch: Dairy #1

Received: 12/13/2023 15:26
Reported: 12/18/2023 15:18

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23L0838-01	Home	Drinking Water	Jim Vieira		12/13/2023 8:00
23L0838-02	Well #1	Drinking Water	Jim Vieira		12/13/2023 8:10
23L0838-03	Well #2	Drinking Water	Jim Vieira		12/13/2023 8:20
23L0838-04	Garet	Drinking Water	Jim Vieira		12/13/2023 8:30
23L0838-05	Grants	Drinking Water	Jim Vieira		12/13/2023 0:00
23L0838-06	Lander	Drinking Water	Jim Vieira		12/13/2023 8:40
23L0838-07	Hirdes	Drinking Water	Jim Vieira		12/13/2023 8:50
23L0838-08	Dads	Drinking Water	Jim Vieira		12/13/2023 9:00
23L0838-09	Verduga	Drinking Water	Jim Vieira		12/13/2023 9:15

Default Cooler Temperature on Receipt °C: 2.4
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

1910 W. McKinley Ave Suite 110 Fresno, CA 93728 559-233-6129 www.dellavallelab.com

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LABORATORY INC

GJC Dairy
P.O. Box 2397
Turlock, CA 95381

Account# 00-0025382
Account Manager: Ben Nydam
Submitted By: F & R Ag Services
Ranch: Dairy #1

Received: 12/13/2023 15:26
Reported: 12/18/2023 15:18

Sample Results

Sample: Home
23L0838-01 (Water)

Sampled: 12/13/2023 8:00
Sampled By: Jim Vieira

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.66**	mmhos/cm	0.01	1		12/14/23 19:18	SM 2510 B		BEL0659
Electrical Conductivity umhos	**665**	umhos/cm	10.0	1		12/14/23 19:18	SM 2510 B		BEL0659
Ammonia (as N)	*	mg/L	0.00	1		12/13/23 08:00	Field		BEL0584
Nitrate Nitrogen as NO3N	**8.4**	mg/L	0.1	1	10	12/14/23 23:25	EPA 300.0		BEL0594
Temperature	**25.0**	units	0.0	1		12/14/23 19:18	SM 4500-H+	H	BEL0659
pH	**7.8**	units	1.0	1		12/14/23 19:18	SM 4500-H+	H	BEL0659

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LABORATORY INC

GJC Dairy
P.O. Box 2397
Turlock, CA 95381

Account# 00-0025382
Account Manager: Ben Nydam
Submitted By: F & R Ag Services
Ranch: Dairy #1

Received: 12/13/2023 15:26
Reported: 12/18/2023 15:18

Sample Results
(Continued)

Sample: Well #1
23L0838-02 (Water)

Sampled: 12/13/2023 8:10
Sampled By: Jim Vieira

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.31**	mmhos/cm	0.01	1		12/14/23 19:19	SM 2510 B		BEL0659
Electrical Conductivity umhos	**1310**	umhos/cm	10.0	1		12/14/23 19:19	SM 2510 B		BEL0659
Ammonia (as N)	*	mg/L	0.00	1		12/13/23 08:10	Field		BEL0584
Nitrate Nitrogen as NO3N	**37.2**	mg/L	0.1	1	10	12/14/23 23:46	EPA 300.0		BEL0594
Temperature	**25.0**	units	0.0	1		12/14/23 19:19	SM 4500-H+	H	BEL0659
pH	**7.5**	units	1.0	1		12/14/23 19:19	SM 4500-H+	H	BEL0659

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LABORATORY INC

GJC Dairy
P.O. Box 2397
Turlock, CA 95381

Account# 00-0025382
Account Manager: Ben Nydam
Submitted By: F & R Ag Services
Ranch: Dairy #1

Received: 12/13/2023 15:26
Reported: 12/18/2023 15:18

Sample Results
(Continued)

Sample: Well #2
23L0838-03 (Water)

Sampled: 12/13/2023 8:20
Sampled By: Jim Vieira

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.44**	mmhos/cm	0.01	1		12/14/23 19:20	SM 2510 B		BEL0659
Electrical Conductivity umhos	**1440**	umhos/cm	10.0	1		12/14/23 19:20	SM 2510 B		BEL0659
Ammonia (as N)	*	mg/L	0.00	1		12/13/23 08:20	Field		BEL0584
Nitrate Nitrogen as NO3N	ND	mg/L	0.1	1	10	12/15/23 00:07	EPA 300.0		BEL0594
Temperature	**25.0**	units	0.0	1		12/14/23 19:20	SM 4500-H+	H	BEL0659
pH	**7.8**	units	1.0	1		12/14/23 19:20	SM 4500-H+	H	BEL0659

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DELLAVALLE™
LABORATORY INC

GJC Dairy
P.O. Box 2397
Turlock, CA 95381

Account# 00-0025382
Account Manager: Ben Nydam
Submitted By: F & R Ag Services
Ranch: Dairy #1

Received: 12/13/2023 15:26
Reported: 12/18/2023 15:18

Sample Results
(Continued)

Sample: Garet
23L0838-04 (Water)

Sampled: 12/13/2023 8:30
Sampled By: Jim Vieira

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**2.63**	mmhos/cm	0.01	1		12/14/23 19:22	SM 2510 B		BEL0659
Electrical Conductivity umhos	**2630**	umhos/cm	10.0	1		12/14/23 19:22	SM 2510 B		BEL0659
Ammonia (as N)	*	mg/L	0.00	1		12/13/23 08:30	Field		BEL0584
Nitrate Nitrogen as NO3N	**124**	mg/L	0.1	1	10	12/15/23 00:28	EPA 300.0		BEL0594
Temperature	**25.0**	units	0.0	1		12/14/23 19:22	SM 4500-H+	H	BEL0659
pH	**7.2**	units	1.0	1		12/14/23 19:22	SM 4500-H+	H	BEL0659

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DELLAVALLE™
LABORATORY INC

GJC Dairy
P.O. Box 2397
Turlock, CA 95381

Account# 00-0025382
Account Manager: Ben Nydam
Submitted By: F & R Ag Services
Ranch: Dairy #1

Received: 12/13/2023 15:26
Reported: 12/18/2023 15:18

Sample Results
(Continued)

Sample: Grants
23L0838-05 (Water)

Sampled: 12/13/2023 0:00
Sampled By: Jim Vieira

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.97**	mmhos/cm	0.01	1		12/14/23 19:23	SM 2510 B		BEL0659
Electrical Conductivity umhos	**974**	umhos/cm	10.0	1		12/14/23 19:23	SM 2510 B		BEL0659
Ammonia (as N)	*	mg/L	0.00	1		12/13/23 00:00	Field		BEL0584
Nitrate Nitrogen as NO3N	**2.9**	mg/L	0.1	1	10	12/15/23 00:48	EPA 300.0		BEL0594
Temperature	**25.0**	units	0.0	1		12/14/23 19:23	SM 4500-H+	H	BEL0659
pH	**7.8**	units	1.0	1		12/14/23 19:23	SM 4500-H+	H	BEL0659

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Turlock, CA 95381

Account# 00-0025382
Account Manager: Ben Nydam
Submitted By: F & R Ag Services
Ranch: Dairy #1

Received: 12/13/2023 15:26
Reported: 12/18/2023 15:18

Sample Results
(Continued)

Sample: Lander
23L0838-06 (Water)

Sampled: 12/13/2023 8:40
Sampled By: Jim Vieira

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.95**	mmhos/cm	0.01	1		12/14/23 19:25	SM 2510 B		BEL0659
Electrical Conductivity umhos	**945**	umhos/cm	10.0	1		12/14/23 19:25	SM 2510 B		BEL0659
Ammonia (as N)	*	mg/L	0.00	1		12/13/23 08:40	Field		BEL0584
Nitrate Nitrogen as NO3N	**32.4**	mg/L	0.1	1	10	12/15/23 01:09	EPA 300.0		BEL0594
Temperature	**25.0**	units	0.0	1		12/14/23 19:25	SM 4500-H+	H	BEL0659
pH	**7.7**	units	1.0	1		12/14/23 19:25	SM 4500-H+	H	BEL0659

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Turlock, CA 95381

Account# 00-0025382
Account Manager: Ben Nydam
Submitted By: F & R Ag Services
Ranch: Dairy #1

Received: 12/13/2023 15:26
Reported: 12/18/2023 15:18

Sample Results
(Continued)

Sample: Hirdes
23L0838-07 (Water)

Sampled: 12/13/2023 8:50
Sampled By: Jim Vieira

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.79**	mmhos/cm	0.01	1		12/14/23 19:26	SM 2510 B		BEL0659
Electrical Conductivity umhos	**790**	umhos/cm	10.0	1		12/14/23 19:26	SM 2510 B		BEL0659
Ammonia (as N)	*	mg/L	0.00	1		12/13/23 08:50	Field		BEL0584
Nitrate Nitrogen as NO3N	**9.2**	mg/L	0.1	1	10	12/15/23 01:30	EPA 300.0		BEL0594
Temperature	**25.0**	units	0.0	1		12/14/23 19:26	SM 4500-H+	H	BEL0659
pH	**7.8**	units	1.0	1		12/14/23 19:26	SM 4500-H+	H	BEL0659

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LABORATORY INC

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Account# 00-0025382
Account Manager: Ben Nydam
Submitted By: F & R Ag Services
Ranch: Dairy #1

Received: 12/13/2023 15:26
Reported: 12/18/2023 15:18

Sample Results
(Continued)

Sample: Dads
23L0838-08 (Water)

Sampled: 12/13/2023 9:00
Sampled By: Jim Vieira

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.71**	mmhos/cm	0.01	1		12/14/23 19:28	SM 2510 B		BEL0659
Electrical Conductivity umhos	**712**	umhos/cm	10.0	1		12/14/23 19:28	SM 2510 B		BEL0659
Ammonia (as N)	*	mg/L	0.00	1		12/13/23 09:00	Field		BEL0584
Nitrate Nitrogen as NO3N	**15.0**	mg/L	0.1	1	10	12/15/23 01:51	EPA 300.0		BEL0594
Temperature	**25.0**	units	0.0	1		12/14/23 19:28	SM 4500-H+	H	BEL0659
pH	**7.8**	units	1.0	1		12/14/23 19:28	SM 4500-H+	H	BEL0659

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LABORATORY INC

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Turlock, CA 95381

Account# 00-0025382
Account Manager: Ben Nydam
Submitted By: F & R Ag Services
Ranch: Dairy #1

Received: 12/13/2023 15:26
Reported: 12/18/2023 15:18

Sample Results
(Continued)

Sample: Verduga
23L0838-09 (Water)

Sampled: 12/13/2023 9:15
Sampled By: Jim Vieira

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.03**	mmhos/cm	0.01	1		12/14/23 19:29	SM 2510 B		BEL0659
Electrical Conductivity umhos	**1030**	umhos/cm	10.0	1		12/14/23 19:29	SM 2510 B		BEL0659
Ammonia (as N)	*	mg/L	0.00	1		12/13/23 09:15	Field		BEL0584
Nitrate Nitrogen as NO3N	**30.2**	mg/L	0.1	1	10	12/15/23 02:11	EPA 300.0		BEL0594
Temperature	**25.0**	units	0.0	1		12/14/23 19:29	SM 4500-H+	H	BEL0659
pH	**7.2**	units	1.0	1		12/14/23 19:29	SM 4500-H+	H	BEL0659

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Submitted By: F & R Ag Services
Ranch: Dairy #1

Received: 12/13/2023 15:26
Reported: 12/18/2023 15:18

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0594										
Blank (BEL0594-BLK1)					Prepared & Analyzed: 12/14/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0594-BLK2)					Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0594-BLK3)					Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0594-BLK4)					Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0594-BLK5)					Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEL0594-BS1)					Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	4.6		0.1	mg/L	5.000		91.6	90-110		
LCS (BEL0594-BS2)					Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	5.2		0.1	mg/L	5.000		105	90-110		
LCS (BEL0594-BS3)					Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	4.6		0.1	mg/L	5.000		92.1	90-110		
LCS (BEL0594-BS4)					Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	4.7		0.1	mg/L	5.000		93.9	90-110		
Duplicate (BEL0594-DUP1)		**Source: 23L0838-01**			Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	8.5		0.1	mg/L		8.4			0.603	10
Duplicate (BEL0594-DUP2)		**Source: 23L0856-02**			Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	2.0		0.1	mg/L		1.9			1.63	10
Duplicate (BEL0594-DUP3)		**Source: 23L0892-03**			Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	4.5		0.1	mg/L		4.6			1.08	10
Duplicate (BEL0594-DUP4)		**Source: 23L0910-05**			Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	8.0		0.1	mg/L		8.0			0.212	10
Matrix Spike (BEL0594-MS1)		**Source: 23L0838-01**			Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	13.8		0.1	mg/L	5.000	8.4	107	90-110		
Matrix Spike (BEL0594-MS2)		**Source: 23L0856-02**			Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	6.8		0.1	mg/L	5.000	1.9	96.3	90-110		

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Account Manager: Ben Nydam
Submitted By: F & R Ag Services
Ranch: Dairy #1

Received: 12/13/2023 15:26
Reported: 12/18/2023 15:18

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0594 (Continued)										
Matrix Spike (BEL0594-MS3)			**Source: 23L0892-03**		Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	9.4		0.1	mg/L	5.000	4.6	95.9	90-110		
Matrix Spike (BEL0594-MS4)			**Source: 23L0910-05**		Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	12.9		0.1	mg/L	5.000	8.0	97.7	90-110		
Reference (BEL0594-SRM1)					Prepared & Analyzed: 12/14/2023					
Nitrate Nitrogen as NO3N	9.2			mg/L	10.00		92.2	90-110		
Reference (BEL0594-SRM2)					Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	9.3			mg/L	10.00		92.7	90-110		
Reference (BEL0594-SRM3)					Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	9.3			mg/L	10.00		93.3	90-110		
Reference (BEL0594-SRM4)					Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	9.4			mg/L	10.00		93.6	90-110		
Reference (BEL0594-SRM5)					Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	9.4			mg/L	10.00		94.1	90-110		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0659										
Blank (BEL0659-BLK1)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
pH	5.5		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Blank (BEL0659-BLK2)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
pH	7.0		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Blank (BEL0659-BLK3)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
pH	7.7		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Duplicate (BEL0659-DUP1)		**Source: 23L0838-09**			Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	1.04		0.01	mmhos/cm		1.03			0.618	10
Electrical Conductivity umhos	1040		10.0	umhos/cm		1030			0.618	10
pH	7.2		1.0	units		7.2			0.695	10
Duplicate (BEL0659-DUP2)		**Source: 23L0852-04**			Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	0.89		0.01	mmhos/cm		0.90			1.22	10
Electrical Conductivity umhos	893		10.0	umhos/cm		904			1.22	10
pH	8.0		1.0	units		8.0			0.500	10
Reference (BEL0659-SRM1)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	429			umhos/cm	426.0		101	90-110		
Reference (BEL0659-SRM2)					Prepared & Analyzed: 12/14/2023					
pH	7.5			units	7.520		100	67021-101.32		
Reference (BEL0659-SRM3)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	1020			umhos/cm	1000		102	90-110		
Electrical Conductivity umhos	1020			umhos/cm	1000		102	90-110		
Reference (BEL0659-SRM4)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	1050			umhos/cm	1000		105	90-110		
Electrical Conductivity umhos	1050			umhos/cm	1000		105	90-110		
Reference (BEL0659-SRM5)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	1030			umhos/cm	1000		103	90-110		

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Received: 12/13/2023 15:26
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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0659 (Continued)										
Reference (BEL0659-SRM5)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity umhos	1030			umhos/cm	1000		103	90-110		
Reference (BEL0659-SRM6)					Prepared & Analyzed: 12/14/2023					
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEL0659-SRM7)					Prepared & Analyzed: 12/14/2023					
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEL0659-SRM8)					Prepared & Analyzed: 12/14/2023					
pH	4.0			units	4.000		100	97.5-102.5		

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12/13/23 15:26 23L0838

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 • 800 228-9896 • Fax 559 268-8174

JG

No. of Water Samples 9 No. of Bottles 9

Water Type: [X] Drinking Water [ ] Wastewater
[ ] Ag Water [ ] Groundwater [ ] Monitoring Well
[ ] Supply Water

Other: No. of Samples **Soil** ____ **Tissue** ____ **Manure** ____

Logged By ____ Results Needed By ____

BILL TO: 25382 / 08
Acct#/Cons. Purchase Order No

GJC DAIRY
Company Name

Address
TURLOCK CA
City, State Zip

Telephone Fax

Cell/Email

REQUESTED BY F&R Ag Services

PROJECT DAIRY #1

Crop ____ Intended ____

Sampled By JIM VIEIRA

Notes: Jamie - 12/18/23 13:17 - no ph # on file for F&R Ag, sent email to Sean to call for results/12/18/23 13:33-Per Sean, OK & thanks

Analysis and Bottles Required:

[X] Well Water **(DWW1)** - EC, NO_3-N, Field NH_4-N, if detected run NH_4-N (1-1 liter plastic, unpreserved) **(Annually)**
[ ] Well Water **(DWW2)** - DWW1 plus HCO_3, CO_3, Cl, SO_4, Dissolved: Ca, Mg, Na, TDS (1-1 liter plastic, unpreserved) **(Every 5 yrs)**
[ ] Process Water **(DPW1)** - EC, NO_3-N, NH_4-N, TKN, TDS, TP, TK (1-1 liter plastic, unpreserved) **(Quarterly)**
[ ] Process Water **(DPW2)** - DPW1 plus CO_3, HCO_3, Cl, SO_4, Total: Ca, Mg, Na, (1-1 liter plastic, unpreserved) **(Every 2 yrs)**
[ ] Canal Water **(DCW1)**-EC, NO_3-N,TKN,TN,TDS (1-1 liter plastic,unpreserved)
[ ] Water – EC, NO_3-N, TDS, TP (1-1 liter plastic, unpreserved)
[ ] Manure **(DM1)** - N, P, K, Moisture **(2x/yr)**
[ ] Manure **(DM2)** - DM1 plus Ash, Ca, Mg, Na, S, Cl **(Every 2 yrs)**
[ ] Compost **(OSA2)** – Moisture, N, P, K, Ash, OM, C:N ratio, Ca, Mg, Na, Fe, Cu, Mn, Zn, S
[ ] Harvest Tissue **(CRA1)** - Moisture , N, P, K, Ash
[ ] Tissue – % Moisture
[ ] Fall Soil – **(DS2)** - FA1 (0-1') **plus CEC, OM**; NO_3-N (1-2')
[ ] OTHER:

COPIES TO: labs@fragservices.com, slocke@fragservices.com
[ ] Co Health Dept
[ ] RWQCB [X] Copy of Chain
[ ] State Forms [ ] QA/QC Documents

Description of Sample	Field NH_4-N	Date/Time Sampled	Rec'd Temp °C	Description of Sample	Field NH_4-N	Date/Time Sampled	Rec'd Temp °C
1. HOME		12/13 8:00	2.4	6. LANDER		12/13 8:40	0.6
2. WELL #1		8:10	5.4	7. HIRDES		8:50	4.0
3. WELL #2		8:20	0.1	8. DADS		9:00	1.0
4. GARET		8:30	-0.7	9. VERDUGA		9:15	1.8
5. GRANTS			0.0	10.			

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First		F+R	12/13 11:00	
Second	JLF	DLV	12-13-23 1:21	12-13-23 3:01
Third				
Fourth		DL	12/13/23 15:26	

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days, overdue accounts will be charged a dated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc (cal). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through *cal* under its Rules and Procedures. The parties will equally bear the costs of mediation arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:

Shipping
$ ____ In
$ ____ Out

Amt Paid ____ Rec By ____ Check # ____ Date ____

Signature

IR Thermometer SN: 200560723
Correction Factor: 0°C
Calibration Due: 03/06/2024
Location: Laboratory

Paste Lat

12/13/23 15:26

23L0838

Shipping Information: Shipped In ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐ Other ☐ _____

☐ Samples refridgerated before pick up | ☐ Picked up samples placed in Ice chest

Container: Ice Chest ☒ Box ☐ None ☐ | **Refrigerant:** Wet Ice ☒ Blue Ice ☐ None ☐

Samples Preserved with HNO_3 or H_2SO_4 were: ☐ Received Preserved ☐ Preserved Upon Receipt at Laboratory

	Type of Container(s) Received	Sample Number 1	2	3	4	5	6	7	8	9	10
	Sample Containers for Internal (DLI) Use *(Containers that go into the Lab)*										
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic	1	1	1	1	1	1	1	1	1	
	250 mL HNO_3 (Red) Plastic										
	* pH Value										
	250 mL H_2SO_4 (Yellow) Plastic										
	* pH Value										
	500 mL unpreserved (White) Plastic										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
Special	500mL unpreserved (White) Glass										
	PO4-P Kit										
	Other:										
	Sample Containers for Subcontracted ("Send Out") Analyses *(Containers that go in the Subcontract ("Send Out") Refrigerator)*										
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	250 mL H_2SO_4 (Yellow) Plastic										
	500 mL HNO_3 (Red)										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
	1 L HNO_3 (Red)										
VOA Vials	40 mL VOA, $Na_2S_2O_3$ + MCAA (EPA531)										
	40 mL VOA, $Na_2S_2O_3$ (EPA547)										
	40mL AG VOA unpreserved (White) (Set of 3)										
	40 mL AG VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
	40mL VOA, H_3PO_4 (Set of 3)										
	40 mL VOA, HCl (Blue) (Set of 3)										
	40 mL VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
Glass	250 mL AG unpreserved (White)										
	250 mL AG H_2SO_4 (Yellow)										
	250 mL AG $Na_2S_2O_3$ (Green)										
	250 mL AG $Na_2S_2O_3$ + MCAA										
	500 mL glass unpreserved (White)										
	500 mL AG HCl (Blue)										
	1 L AG unpreserved (White)										
	1 L AG H_2SO_4 (Yellow)										
	1 L AG $Na_2S_2O_3$ (Green)										
	1 L AG HCl (Blue)										
Special	Cr^{6+} - 50mL Plastic w/Borate/HCO_3/CO_3										
	Cyanide - 500 mL NaOH										
	Asbestos - 1L P wrapped in foil (Set of 2)										
	Sulfide - 1 L **AG** or **P** NaOH + ZnAc										
	Chlorite/Bromate - 250 mL AG with EDA										
	HAA5 - 250mL AG Ammonium Chlorite										
	DO KIT										
	Other:										
	Other:										

SEND OUT